

Part A: Fundamentals — Answers

1. What is Data Analysis?

Ans. Data Analysis is the process of collecting, cleaning, transforming, exploring, and interpreting data to find useful information, patterns, and insights. It helps organizations make better decisions based on data.

Main steps of Data Analysis:

1. Collect data from different sources.
2. Clean the data by handling missing values, duplicates, and errors.
3. Transform and organize the data.
4. Perform exploratory data analysis (EDA).
5. Visualize the data using graphs and charts.
6. Identify patterns and relationships.
7. Draw conclusions and communicate insights.

2. How to Plan a Data Science Project

Ans. A Data Science project can be planned using the following steps:

1. Define the Problem
Clearly identify the business problem and objective.
2. Collect Data
Gather relevant data from sources such as CSV files, JSON, SQL databases, APIs, etc.
3. Understand the Data
Examine columns, data types, number of records, and relationships between variables.
4. Clean the Data
Handle missing values, duplicates, incorrect values, and outliers.
5. Exploratory Data Analysis (EDA)
Use statistics and visualizations to discover patterns, trends, and relationships.
6. Feature Engineering
Create or modify features that can improve the performance of the model.
7. Data Preprocessing
Perform encoding, scaling, normalization, and other transformations required by ML algorithms.

8. Split the Data
Divide the dataset into training and testing **sets**.
9. Select and Train a Model
Choose a suitable machine learning algorithm and train it using the training data.
10. Evaluate the Model
Measure performance using appropriate metrics such as accuracy, precision, recall, F1-score, or ROC-AUC.
11. Tune the Model
Optimize hyperparameters to improve model performance.
12. Deploy the Model
Make the trained model available for real-world use.
13. Monitor and Maintain
Continuously monitor model performance and update the model when necessary.

Flow:

Problem Definition → Data Collection → Data Understanding → Data Cleaning → EDA → Feature Engineering → Preprocessing → Model Training → Evaluation → Deployment → Monitoring

3. Machine Learning Problem Statement: Customer Churn

Problem Statement:

Develop a machine learning classification model that predicts whether a customer will churn (leave the company) or not based on their purchase behavior and other relevant customer attributes.

Ans. Target Variable:

- Churn = 1 → Customer will churn
- Churn = 0 → Customer will not churn

Possible input features:

- Purchase frequency
- Total purchase amount
- Number of purchases
- Annual income
- Customer age
- Customer tenure

- Average transaction value
- Last purchase date
- Customer complaints/interactions

Type of ML Problem:

This is a supervised learning → binary classification problem because the target has two possible outcomes: Churn / No Churn.

Goal:

Identify customers who are likely to churn so that the company can take preventive actions such as offering discounts, personalized offers, or improving customer service.

Q4. What are Tensors?

Ans. A tensor is a mathematical object used to store and represent data in multiple dimensions. In programming and machine learning, tensors are commonly used to store numerical data such as images, text representations, audio, and datasets.

You can think of a tensor as a generalized array:

Tensor	Dimension	Example
Scalar	0D	5
Vector	1D	[1, 2, 3]
Matrix	2D	[[1,2], [3,4]]
3D Tensor	3D	Collection of matrices
4D Tensor	4D	Collection of 3D tensors

Tensors using NumPy

NumPy uses ndarray to represent multidimensional arrays, which can be used as tensors.

First, import NumPy:

```
import numpy as np
```

1. Scalar — 0D Tensor

A scalar contains only one value.

```
a = np.array(5)
```

```
print(a)
print(a.ndim)
print(a.shape)
```

Output:

```
5
0
()
```

Here:

- `ndim = 0` → 0 dimensions
- `shape = ()` → no dimensions

2. Vector — 1D Tensor

A vector is a one-dimensional array.

```
a = np.array([10, 20, 30, 40])
```

```
print(a)
print(a.ndim)
print(a.shape)
```

Output:

```
[10 20 30 40]
1
(4,)
```

It contains 4 elements, so its shape is (4,).

3. Matrix — 2D Tensor

A matrix contains rows and columns.

```
a = np.array([
    [1, 2, 3],
    [4, 5, 6]
```

```
)
```

```
print(a)
```

```
print(a.ndim)
```

```
print(a.shape)
```

Output:

```
[[1 2 3]
```

```
 [4 5 6]]
```

```
2
```

```
(2, 3)
```

This tensor has:

- 2 rows
- 3 columns
- 2 dimensions

So its shape is (2, 3).

4. 3D Tensor

A 3D tensor can be thought of as a collection of matrices.

```
a = np.array([  
    [[1, 2], [3, 4]],  
    [[5, 6], [7, 8]]
```

```
])
```

```
print(a)
```

```
print(a.ndim)
```

```
print(a.shape)
```

Output:

```
[[[1 2]
```

```
 [3 4]]
```

[[5 6]
[7 8]]]

3

(2, 2, 2)

Here the shape (2, 2, 2) means:

2 matrices $\times$ 2 rows $\times$ 2 columns